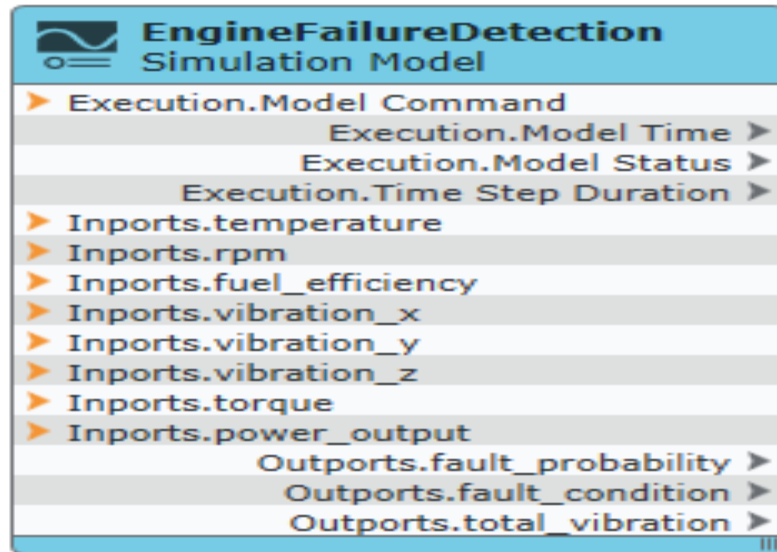


IVHM Test Cases Demo Using NI (Veristand)



Engine failure Detection Demo using NI Veristand



EngineFailureDetection

Execution

Model Command

Start

Model Time

74.68 sec

Model Status

Running

Time Step Duration

20 usec

Inports

temperature

0

rpm

0

fuel_efficiency

0

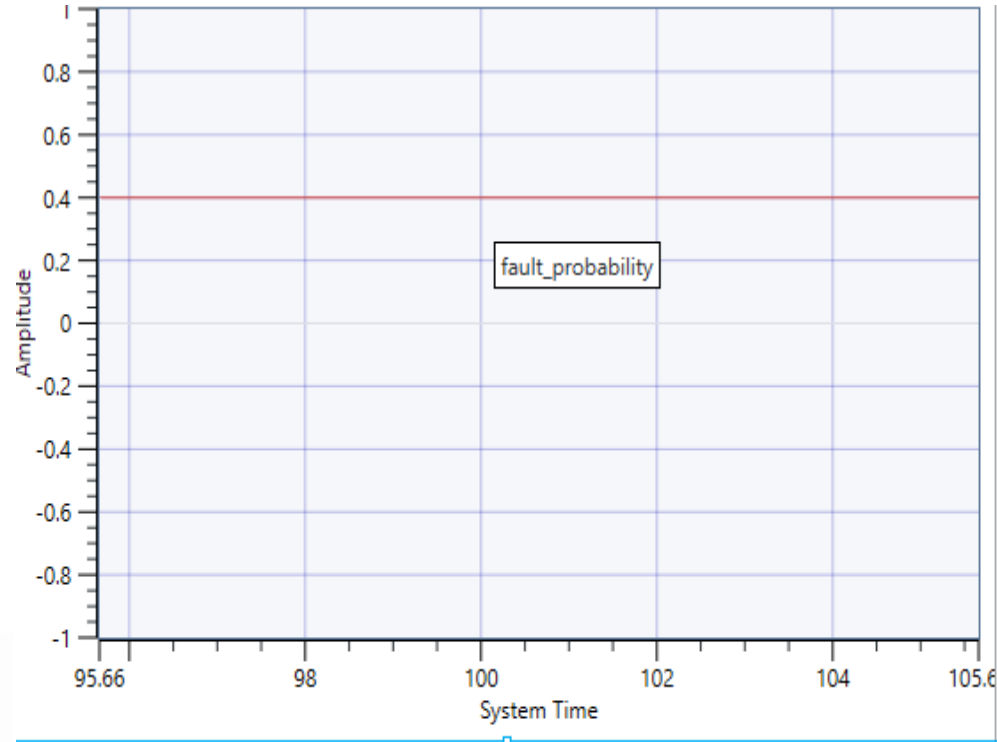
vibration_x

0

vibration_y

0

Engine failure Detection Demo using NI Veristand



EngineFailureDetection

Execution

Model Command:

Model Time:

Model Status:

Time Step Duration:

Inputs

temperature:

rpm:

fuel_efficiency:

vibration_x:

vibration_y:

vibration_z:

torque:

power_output:

Outputs

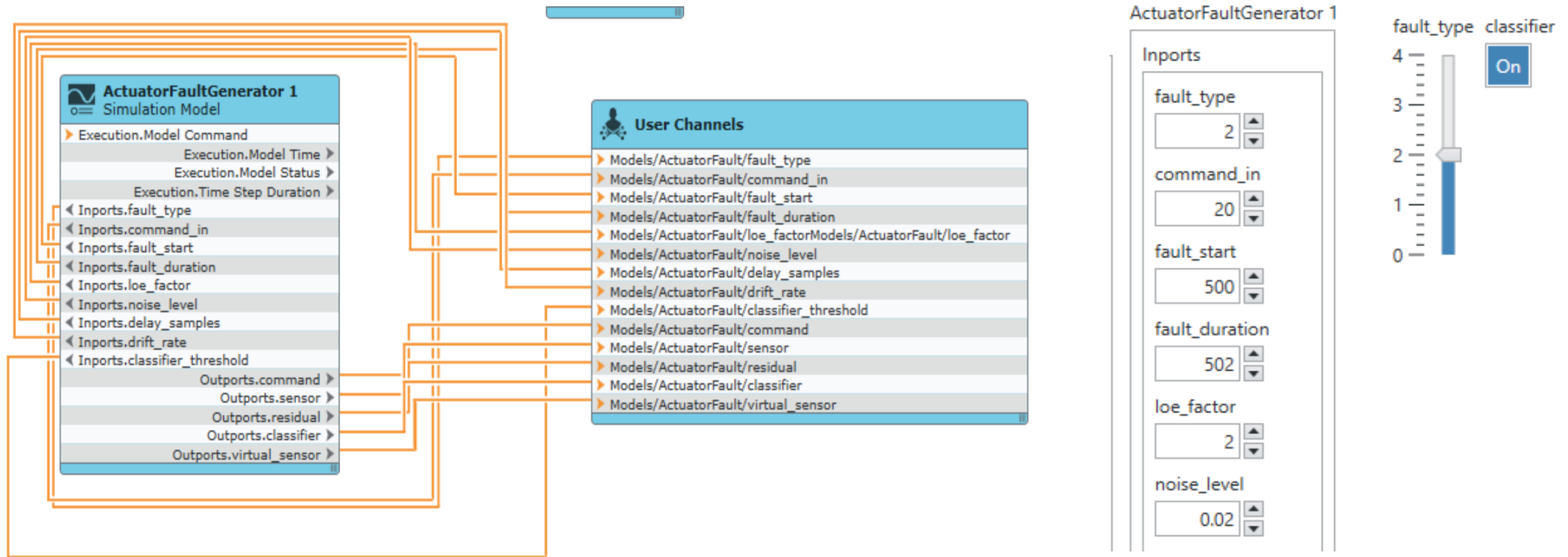
fault_probability:

fault_condition:

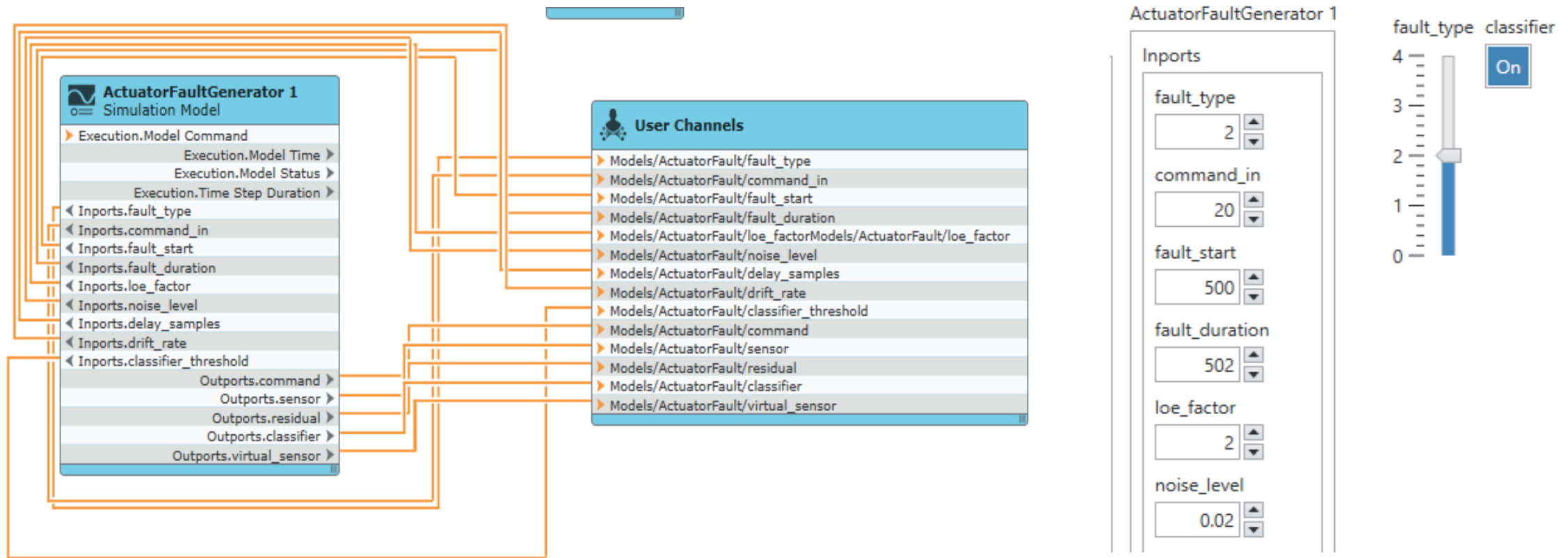
total_vibration:



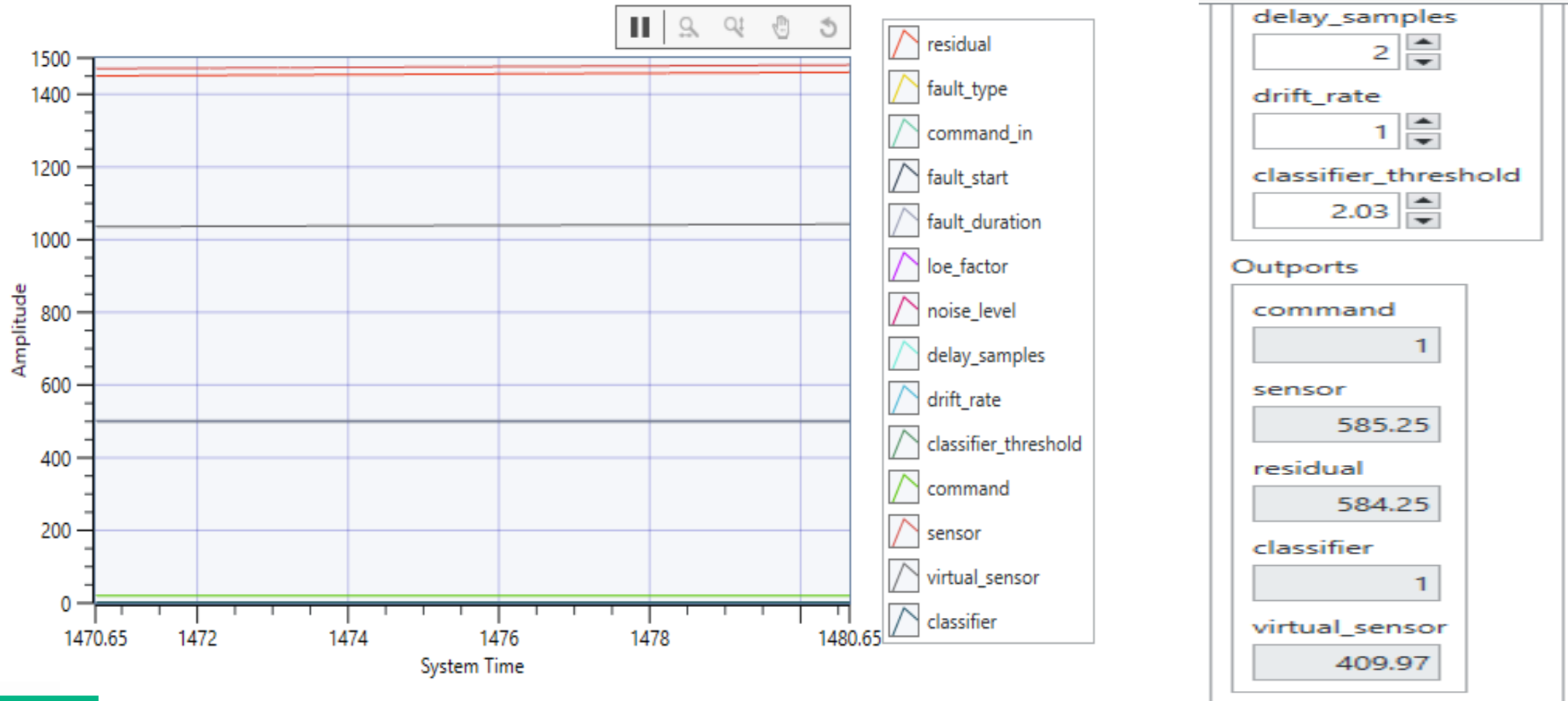
Sensor faults Generator Demo using NI Veristand



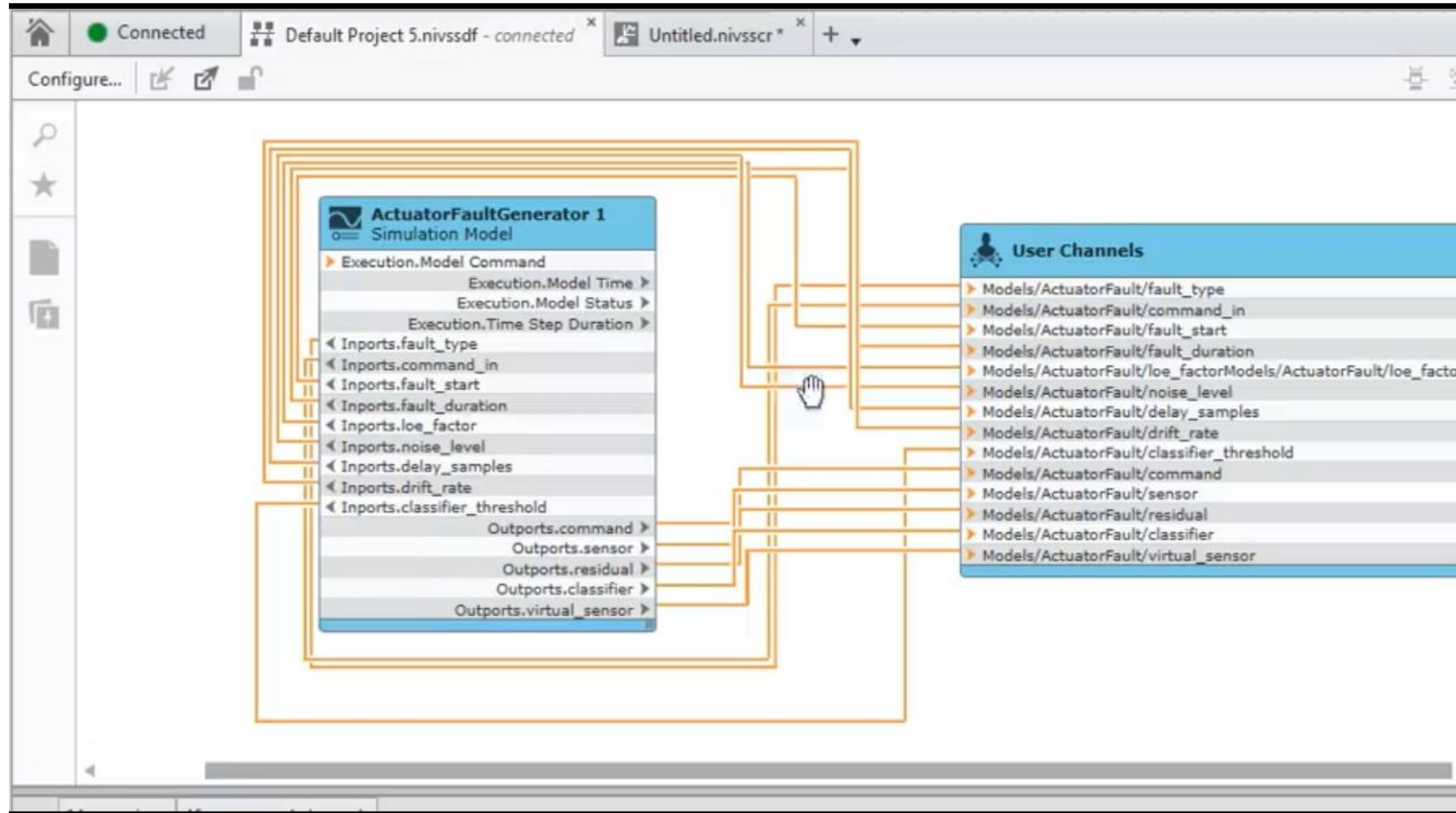
Sensor faults Generator Demo using NI Veristand



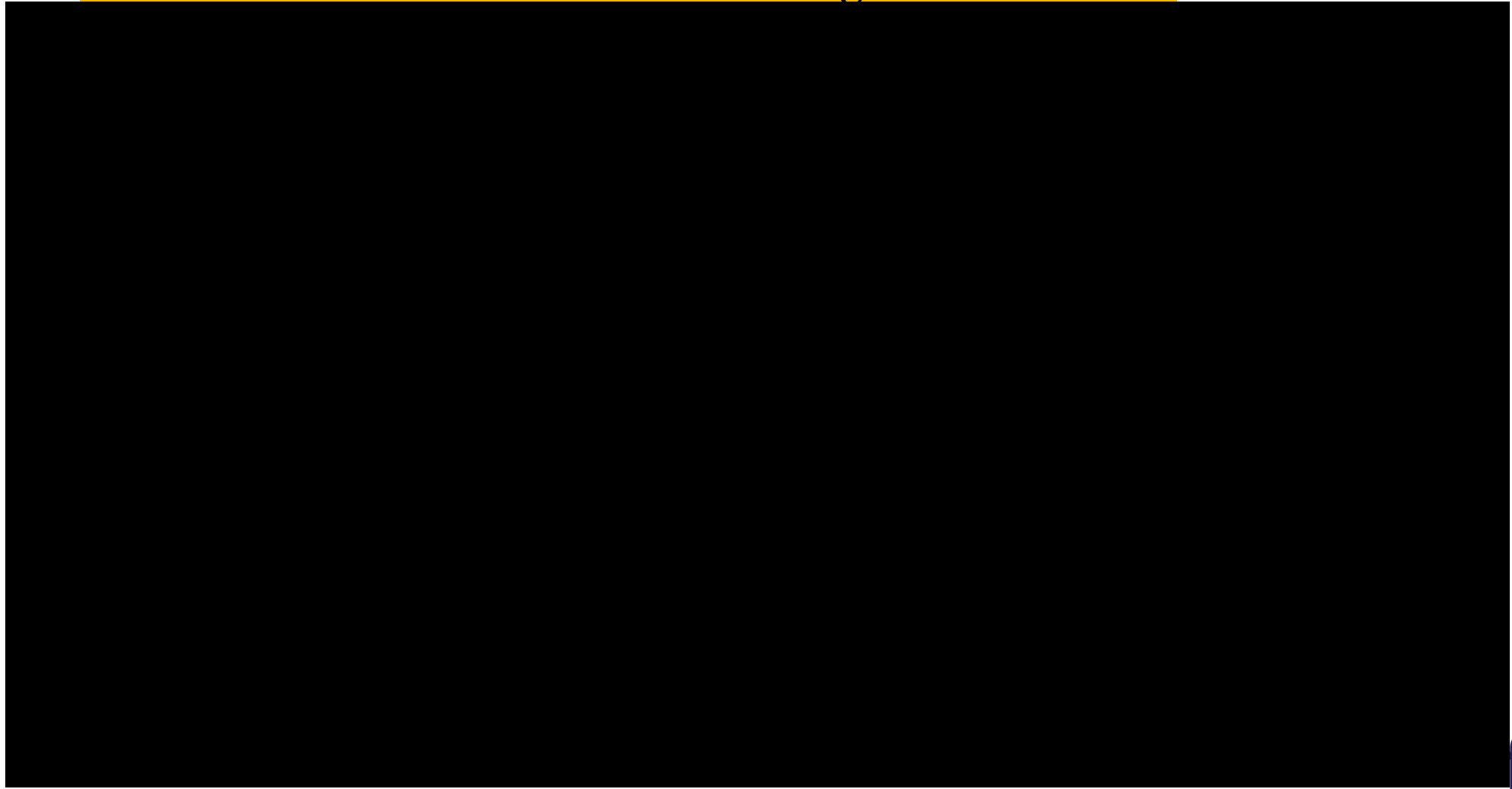
Sensor faults Generator Demo using NI Veristand



Sensor faults Generator Demo using NI Veristand



Sensor faults Generator Demo using NI Veristand

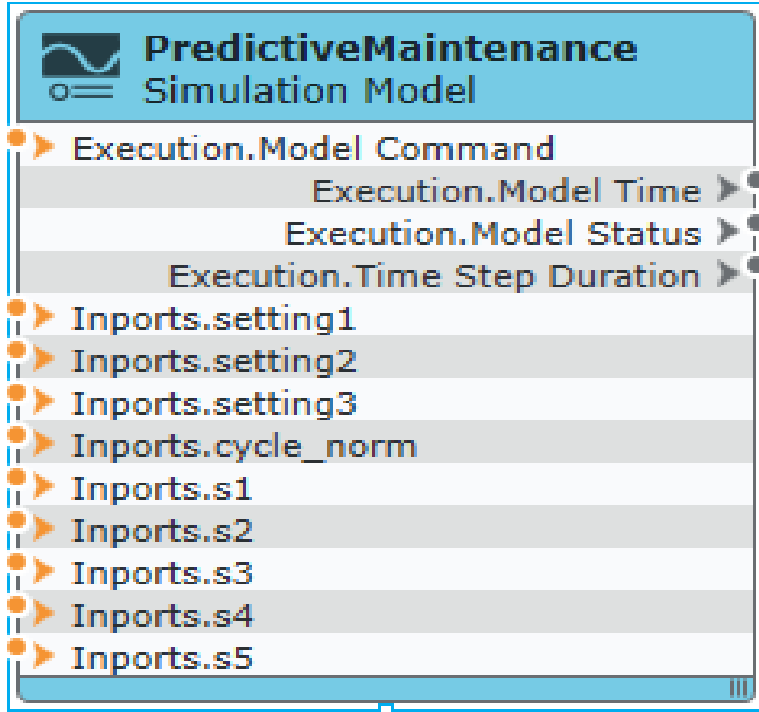


Sensor faults Generator Demo using NI Veristand

```
fault_type = 0  
command_in = 15  
fault_start = 0  
fault_duration = 0  
loe_factor = 0.0  
noise_level = 0.01  
delay_samples = 2  
drift_rate = 0.0  
classifier_threshold = 0.01 # initial; recompute after baseline run
```



Aircraft Predictive Maintenance Demo using NI Veristand



PredictiveMaintenance
Simulation Model

- ▶ Execution.Model Command
 - Execution.Model Time
 - Execution.Model Status
 - Execution.Time Step Duration
- ▶ Inports.setting1
- ▶ Inports.setting2
- ▶ Inports.setting3
- ▶ Inports.cycle_norm
- ▶ Inports.s1
- ▶ Inports.s2
- ▶ Inports.s3
- ▶ Inports.s4
- ▶ Inports.s5

PredictiveMaintenance

Execution

Model Command

Start

Model Time

117.40

Model Status

Running

Time Step Duration

31 use



Aircraft Predictive Maintenance Demo using NI Veristand

